SUSTAINABLE MATERIALS TECHNOLOGY

Green materials: can engineering save its own carbon footprint?

Joshua Otaigbe · October 15, 2026

Steel and cement together account for roughly 15% of global CO2 emissions.

Not transport. Not power generation. The materials that hold buildings up and connect them to roads.

This gets overlooked in most conversations about decarbonization, which tend to focus on what happens after materials are made: cleaner cars, greener electricity grids, better insulation. The production side, the smelting and kilning and chemical reduction that turns raw earth into structural material, gets far less attention than it deserves.

That's starting to change. The race to decarbonize materials production is one of the more consequential engineering stories of the decade. And it's significantly more complicated than swapping fossil fuels for renewables.

Why materials are hard to decarbonize

Burning coal to generate electricity is dirty, but the carbon problem is entirely in the combustion. Replace coal with wind or solar and the problem goes away.

Cement doesn't work that way.

Producing Portland cement requires heating limestone (calcium carbonate) to around 1450°C, which drives off CO2 as a chemical byproduct. About 60% of cement's carbon emissions come from this reaction, not from the fuel used to heat the kiln. Switching to a renewable energy source eliminates the other 40%. The process emissions remain.

Steel has a different but equally stubborn problem. The dominant steelmaking route, basic oxygen furnace production, uses coke both to heat the iron ore and to chemically reduce it, stripping oxygen atoms off iron oxide. Carbon is doing chemistry, not just generating heat.

These are process emissions. They're built into the chemistry of the materials themselves. Getting

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rid of them requires either changing the chemistry or capturing the CO2 at the source. Both are being actively pursued. Neither is easy.

Green steel: how close is it?

Two approaches are serious.

Hydrogen direct reduction (H-DRI)

Replace the coke with hydrogen. When hydrogen reacts with iron ore, the byproduct is water, not CO2. The chemistry works. SSAB, the Swedish steelmaker, produced the world's first hydrogen-reduced steel in 2021 through its HYBRIT joint venture with LKAB and Vattenfall. Volvo received the first commercial delivery. SSAB plans to phase out its coal-based blast furnaces entirely by 2030.

The catch: green hydrogen currently costs 3 to 5 times more than coal-based reduction agents. And the HYBRIT process needs roughly 55 kilowatt-hours of electricity per kilogram of hydrogen produced. At current renewable electricity prices and production volumes, hydrogen-reduced steel costs significantly more than conventional steel. That premium is falling as electrolyzer costs drop and renewable electricity gets cheaper. How fast it falls is genuinely uncertain.

Electric arc furnace (EAF) with scrap

EAFs melt steel scrap using electricity. If that electricity is renewable, the process is very low carbon. EAFs already produce about 30% of global steel, mostly in the US and EU where scrap is abundant. Expanding EAF production is limited by scrap availability: you can only recycle steel that's already been made. As global steel demand grows, particularly in India and Southeast Asia, virgin iron production can't be fully replaced by recycling. H-DRI and EAF are likely to work together: H-DRI produces high-quality "green pig iron" that feeds EAFs, reducing the quality limitations of scrap-only production.

ArcelorMittal, Thyssenkrupp, and BlueScope all have active hydrogen steelmaking programs. The Boston Metal company is pursuing an entirely different route: molten oxide electrolysis, which uses electricity to reduce iron ore directly without any hydrogen or carbon. They raised $262 million in 2022. Their process is earlier stage but avoids hydrogen's infrastructure challenges entirely.

Low-carbon cement: where the progress is

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Supplementary cementitious materials (SCMs)

Portland cement clinker (the CO2-intensive component) can be partially replaced by industrial byproducts that have cementitious properties: fly ash from coal power plants, blast furnace slag from steelmaking, and calcined clays. Replacing 30-50% of clinker with these materials cuts emissions proportionally. The problem: fly ash supply is falling as coal plants close. Slag supply is limited by steel production volumes. Calcined clay is the most abundant SCM and the most scalable. LC3 cement (limestone calcined clay cement), developed partly at EPFL in Switzerland, can replace 50% of clinker using kaolinite clay, a material available almost everywhere. Cuba, India, and Colombia are deploying LC3 at commercial scale.

Novel cement chemistries

Calix, an Australian company, has developed a process called LEILAC that captures process CO2 directly at the kiln. Solidia Technologies produces a cement that cures by absorbing CO2 rather than releasing it. Brimstone Energy has developed a process using calcium silicate rocks instead of limestone, eliminating process emissions entirely. They raised $55 million in 2022 and are building a pilot plant.

None of these are at full commercial scale. All of them are moving faster than cement chemistry has moved in the past 50 years.

Carbon capture on conventional kilns

The cement industry is also pursuing post-combustion carbon capture as a near-term bridge: fit existing kilns with amine scrubbing or oxyfuel combustion and capture the CO2 before it reaches the atmosphere. Expensive. Energy-intensive. But proven technology that can be retrofitted. Heidelberg Materials is building what it claims will be the world's first full-scale carbon capture cement plant in Brevik, Norway, targeting operation in 2024.

Bio-based materials: growing the alternative

Mass timber and engineered wood

Cross-laminated timber (CLT) can replace concrete and steel in multi-story buildings. Trees absorb CO2 as they grow, and that carbon stays locked in the wood for the life of the building. The lifecycle emissions of CLT buildings are substantially lower than concrete equivalents, typically 50-80% lower on a cradle-to-gate basis. Mjøstårnet in Norway reaches 85.4 metres. The carbon savings

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are real, but timber production at the scale needed to substitute meaningfully for concrete globally would require a significant expansion of managed forests.

Mycelium composites

Grow fungal mycelium on agricultural waste, shape it in a mold, then dry it. The result is a rigid, lightweight composite that's fully biodegradable. Ecovative Design has been selling mycelium packaging since 2010 and expanded into construction panels and acoustic insulation. IKEA switched to mycelium packaging for some products. The material is genuinely useful for low-load applications. Its structural properties aren't close to concrete or steel and probably won't be.

Hemp and flax fiber composites

Natural plant fiber composites have been used in automotive interiors for years: BMW, Mercedes, and Audi all use flax or hemp fiber reinforced panels in door cards and trunk liners. The fiber grows fast, sequesters carbon during its lifecycle, and produces a part that's lighter than a glass fiber equivalent. Structural performance is lower, which limits applications to non-critical panels. But the market exists and is growing.

Bamboo

Engineering-grade bamboo products (parallel strand bamboo, laminated bamboo lumber) rival timber in structural performance and have much faster growth cycles: 5 years vs 25-80 years for commercial timber species. MOSO International and Lamboo produce structural bamboo products sold in construction markets. The carbon story is strong. The manufacturing base is still small and geographically concentrated in Asia.

The lifecycle analysis problem

All "green" materials claim needs to be interrogated with lifecycle analysis (LCA).

A material with low production emissions might have high transport emissions if it's shipped across the world. A material that sequesters carbon during its life might release it all at end of life if it's landfilled or incinerated.

Mycelium packaging is fully compostable. But if the composting infrastructure doesn't exist in the region where it's disposed, it ends up in landfill where it generates methane as it breaks down, which is a worse greenhouse gas than CO2 on a 20-year timeframe.

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The honest answer is that LCA is genuinely hard, context-dependent, and frequently abused in marketing materials. A published lifecycle analysis funded by the industry whose product it's evaluating deserves scrutiny. Third-party verified EPDs (Environmental Product Declarations) are the more reliable standard.

Who's investing, and what it signals

ArcelorMittal committed $1 billion to green steel in 2021. Thyssenkrupp's hydrogen steelmaking program received 2 billion in EU and German government support in 2023. The US Inflation Reduction Act extended the 45Q tax credit for carbon capture, worth up to $85 per tonne of CO2 sequestered, which meaningfully improves the economics of carbon capture on cement kilns and steel plants.

On the bio side, Ecovative raised $60 million in 2022. Timber construction is seeing record investment in manufacturing capacity across the US, Europe, and Australia, driven partly by building code changes that now permit mass timber structures above 18 stories in several jurisdictions.

Green steel and low-carbon cement are receiving large industrial capital commitments because the markets are enormous and the incumbent producers have to move. Bio-based materials are receiving venture capital at early scale because the markets are smaller but the margins are potentially higher and the competition from incumbents is lower.

Both are genuine. The timelines are different.

Where this lands

The decarbonization of materials production is probably 20-30 years away from completion at scale. Green hydrogen costs need to fall significantly. Carbon capture needs to scale from demonstration plants to global deployment. Managed forests need time to grow. Bio-based products need supply chains that don't yet exist at anywhere near the required size.

None of that means it's not happening. The direction is clear. The pace is faster than it was 5 years ago. The capital commitment is real.

The engineering challenge of decarbonizing materials production is arguably harder than decarbonizing electricity generation because the process emissions can't be solved by switching fuels. But harder problems have a way of attracting the people who find easier problems boring.

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